Project 605 ⇒ Clickhouse Tutorial

→ We are using clickhouse as the OLAP engine on e-commerce event-logs.

→ Hypothesis: using the first N=5 events in a user_session, we can predict whether the session will have a purchase later.

⇒ ecomm (database)

⇒ events_raw (table)

- → Row ⇒ 1 event
- → First table from the csv
- → Columns: event_time (String)
  - event_type (String)
  - product_id (int)
  - category_id (int)
  - category_code (String) Nullable
  - brand (String) Nullable
  - price (float)
  - user_id (int)
  - user_sessions (String)
- → Engine: Merge Tree
- → Order by: (user_session)

⇒ events_typed (table)

- → Row ⇒ 1 clean event
- → Main analytics table
- → columns same as event_raw + event_time_dt - parsed timestamp
  - event_date - Date
- → Engine: Merge Tree
- → Partition By: event_date
- → Order By: (user_session, event_time_dt)
- → 'category code / brand' are converted to NULL

⇒ session_outcomes (table)

- → Row → 1 user_session
- → Define the supervised label: has_purchase_in_session
- → Group: 'user_session'

⇒ mv_events_raw_to_typed (view)

- → raw → typed automatic pipeline
  - ↳ inserts once into events_raw typed data appears in events_typed

⇒ session_user_history

- → Row: 1 session_outcomes that include per-user history up to that session.
- → This adds the user-history into the current session
- → Columns: prior_sessions
  - prior_purchase_count ⇒ count of past-sessions with a purchase
  - sec_since_last_session ⇒ gap from previous session start
  - hist_conversion_rate ⇒ prior purchase rate over prior sessions
- → Engine: MergeTree
- → Order By: (user_id, session_start, user_session)

⇒ session_prefix_features_n5 (Table)

- → Row: 1 user_session that has exactly N=5 earliest events
- → Turn the first 5 events in time order into session-level numeric features.
- → It performs by using the subquery prefix [order by user_session, event_time_dt then Limit 5 by user_session & user_session ≠ ''
- → Important columns: prefix events [must be 5]
  - prefix views/carts/purchases → counts of each event-type
  - prefix_uniq_products → distinct # product_id
  - prefix_uniq_categories → distinct # category_id
  - prefix_uniq_brands → Distinct non-empty brand
  - (in prefix)
  - prefix avg. price → Avg. price over 5 rows
  - prefix_max price → Max price in prefix
  - prefix_cart_value_sum → sum of price where event_type = 'cart' in prefix
- → Engine: Merge Tree
- → Order by: User_session

⇒ training_sessions_n5

- → Row: user_session that appears in both 'session_prefix_features_n5' & 'session_user_history'
- → ML ready table: early current session behaviours + User_history + output: has_purchase_in_session
- → Columns: All columns of session_prefix_features_n5
  - \+
  - All columns of user_history
  - \+
  - 'has_purchase_in_session'
- → Engine: Merge Tree

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⇒ Session_purchase_predictions

- → Output table: Total rows: 50,000 (test)
- → columns: model name (String)
  - user_session (String)
  - p_purchase (float)
  - pred_purchase (int)
- → Engine: Merge Tree
- → Order By (model_name, user_session)